

Simultaneous Equation Model

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1 Introduction

- We will consider econometric models for data that are jointly determined by two or more economic relations
- These simultaneous equations models differ from those previously studied because in each model there are two or more dependent variables rather than just one
- Simultaneous equations models also differ from most of the econometric models we have considered so far, because they consist of a set of equations

2 A Supply and Demand Model

- A very simple supply and demand model might look like:

$$\text{Demand: } Q = \alpha_1 P + \alpha_2 X + e_d \quad (1)$$

$$\text{Supply: } Q = \beta_1 P + e_s \quad (2)$$

- It takes two equations to describe the supply and demand equilibrium
- The two equilibrium values, for price and quantity, P^* and Q^* , respectively, are determined at the same time

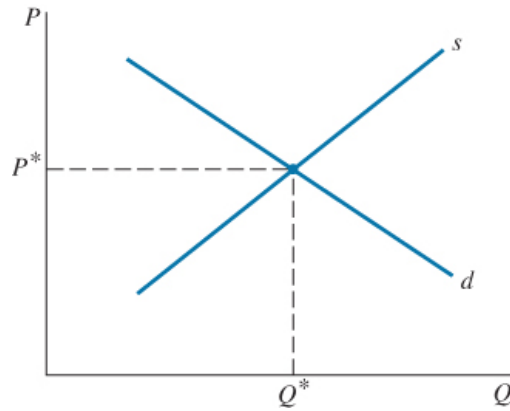


Figure 1: Supply and Demand Equilibrium

- In this model the variables P and Q are called *endogenous variables* because their values are determined within the system we have created
- The endogenous variables P and Q are dependent variables and both are random variables
- The income variable X has a value that is determined outside this system
- Such variables are said to be *exogenous*, and these variables are treated like usual “ x ” explanatory variables
- For the error terms, we have:

$$\begin{aligned} E(e_d) &= 0, & \text{Var}(e_d) &= \sigma_d^2 \\ E(e_s) &= 0, & \text{Var}(e_s) &= \sigma_s^2 \\ \text{Cov}(e_d, e_s) &= 0 \end{aligned} \tag{3}$$

- An “*influence diagram*” is a graphical representation of relationships between the model’s components

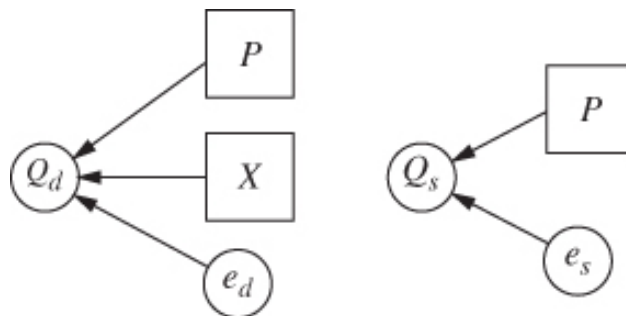


Figure 2: Influence Diagrams for two Regression Models

- The fact that P is an endogenous variable on the right-hand side of the supply and demand equations means that we have an explanatory variable that is random

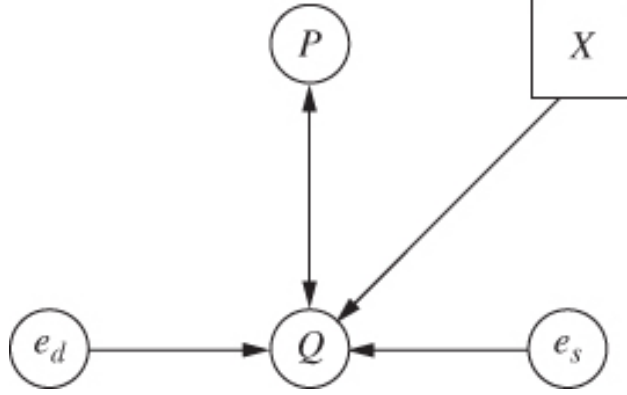


Figure 3: Influence Diagram for a Simultaneous Equations Model

- This is contrary to the usual assumption of “fixed explanatory variables”
- The problem is that the endogenous regressor P is correlated with the random errors, e_d and e_s , which has a devastating impact on our usual least squares estimation procedure, making the least squares estimator biased and inconsistent

3 The Reduced-Form Equations

- The two structural equations Eqs. 1 and 2 can be solved to express the endogenous variables P and Q as functions of the exogenous variable X .
- This reformulation of the model is called the *reduced-form* of the structural equation system
- To solve for P , set Q in the demand and supply equations to be equal:

$$\beta_1 P + e_s = \alpha_1 P + \alpha_2 X + e_d$$

- Solve for P :

$$\begin{aligned} P &= \frac{\alpha_2}{(\beta_1 - \alpha_1)} X + \frac{e_d - e_s}{(\beta_1 - \alpha_1)} \\ &= \pi_1 X + v_1 \end{aligned} \tag{4}$$

- Solving for Q :

$$\begin{aligned} Q &= \beta_1 P + e_s \\ &= \beta_1 \left[\frac{\alpha_2}{(\beta_1 - \alpha_1)} X + \frac{e_d - e_s}{(\beta_1 - \alpha_1)} \right] + e_s \\ &= \frac{\beta_1 \alpha_2}{(\beta_1 - \alpha_1)} X + \frac{\beta_1 e_d - \alpha_1 e_s}{(\beta_1 - \alpha_1)} + e_s \\ &= \pi_2 X + v_2 \end{aligned} \tag{5}$$

- The parameters π_1 and π_2 in Eqs. 4 and 5 are called *reduced-form parameters*. The error terms v_1 and v_2 are called *reduced-form errors*

R Example

```
library(systemfit)
# For details of data see in this example, see the Kmenta, 1986.
data("Kmenta")
attach(Kmenta)
names(Kmenta)

## [1] "consump"    "price"      "income"     "farmPrice" "trend"

# consump = food consumption per capita
# price = ratio of food prices to general consumer prices
# income = disposable income

# Equations for Demand and Supply Respectively:
# Demand:  $Q = P + X$ ,  $X = income$ 
# Supply:  $Q = P$ 

# Reduced-Form Equations (1st Stage Equations)
# for Demand and Supply Respectively:
Q.reduced <- lm(consump~income)
P.reduced <- lm(price~income)
summary(Q.reduced)

##
## Call:
## lm(formula = consump ~ income)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.5118 -2.0608  0.0495  1.6410  4.0363
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  77.01461    4.67944  16.458 2.70e-12 ***
## income        0.24487    0.04765   5.139 6.87e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.457 on 18 degrees of freedom
## Multiple R-squared:  0.5947, Adjusted R-squared:  0.5722
## F-statistic: 26.41 on 1 and 18 DF,  p-value: 6.872e-05
summary(P.reduced)

##
## Call:
## lm(formula = price ~ income)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.199  -2.823   1.268   3.796   6.560
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  72.33922    9.55533   7.571 5.33e-07 ***
```

```
## income      0.28379    0.09729    2.917    0.0092 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.017 on 18 degrees of freedom
## Multiple R-squared:  0.321, Adjusted R-squared:  0.2833
## F-statistic: 8.509 on 1 and 18 DF,  p-value: 0.009202
```

*# Interpretation: beta_income= is the estimated effect of a change
in income (exogenous variable) on the equilibrium price and
quantity (endogenous variables).*

4 The Failure of Least Squares Estimation

- The least squares estimator of parameters in a structural simultaneous equation is biased and inconsistent because of the correlation between the random error and the endogenous variables on the right-hand side of the equation

4.1 The Identification Problem

- In the supply and demand model given by Eqs. 1 and 2:
 - The parameters of the demand equation, α_1 and α_2 , cannot be consistently estimated by any estimation method
 - The slope of the supply equation, β_1 , can be consistently estimated

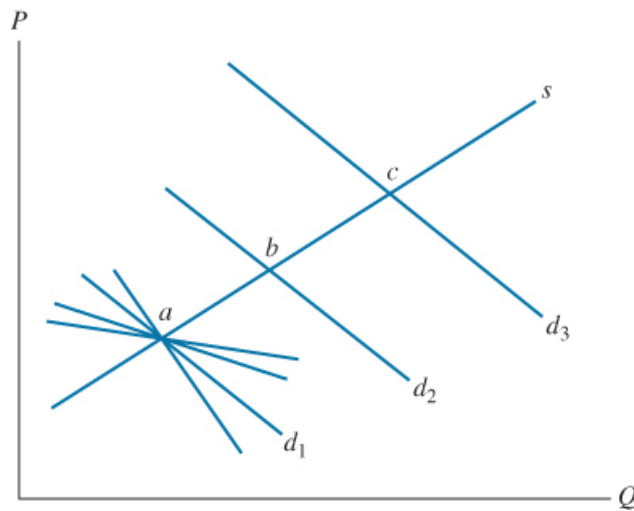


Figure 4: The Effect of Changing Income

- It is the *absence* of variables in one equation that are *present* in another equation that makes parameter estimation possible

4.2 A Necessary Condition for Identification:

A general rule, which is called a *necessary condition* for identification of an equation, is:

- In a system of M simultaneous equations, which jointly determine the values of M endogenous variables, at least $M - 1$ variables must be absent from an equation for estimation of its parameters to be possible
 - When estimation of an equation's parameters is possible, then the equation is said to be identified, and its parameters can be estimated consistently.
 - If fewer than $M - 1$ variables are omitted from an equation, then it is said to be unidentified, and its parameters cannot be consistently estimated
- The identification condition must be checked before trying to estimate an equation
- If an equation is not identified, then changing the model must be considered before it is estimated

5 Two-Stage Least Squares Estimation

- The most widely used method for estimating the parameters of an identified structural equation is called two-stage least squares
- This is often abbreviated as *2SLS*
- The name comes from the fact that it can be calculated using two least squares regressions

Consider the supply equation discussed previously

- We cannot apply the usual least squares procedure to estimate β_1 in this equation because the endogenous variable P on the right-hand side of the equation is correlated with the error term e_s .
- The reduced-form model is:

$$\begin{aligned} P &= E(P) + v_1 \\ &= \pi_1 X + v_1 \end{aligned} \tag{6}$$

- Suppose we know π_1 .
- Then through substitution:

$$\begin{aligned} Q &= \beta_1 [E(P) + v_1] + e_s \\ &= \beta_1 E(P) + (\beta_1 v_1 + e_s) \end{aligned} \tag{7}$$

- We can estimate π_1 using from the reduced-form equation for P
- A consistent estimator for $E(P)$ is:

$$\hat{P} = \hat{\pi}_1 X$$

- Then:

$$Q = \beta_1 \hat{P} + e_s \tag{8}$$

- Estimating Eq. 8 by least squares generates the so-called *two-stage least squares estimator* for β_1 , which is consistent and normally distributed in large samples

- The two stages of the estimation procedure are:
 1. Least squares estimation of the reduced-form equation for P and the calculation of its predicted value $\hat{P}$
 2. Least squares estimation of the structural equation in which the right-hand-side endogenous variable P is replaced by its predicted value $\hat{P}$

- Suppose the first structural equation in a system of M simultaneous equations is:

$$y_1 = \alpha_2 y_2 + \alpha_3 y_3 + \beta_1 x_1 + \beta_2 x_2 + e_1 \quad (9)$$

- If this equation is identified, then its parameters can be estimated in the two steps:
- **Step 1:** Estimate the parameters of the reduced-form equations

$$\begin{aligned} y_2 &= \pi_{12}x_1 + \pi_{22}x_2 + \cdots + \pi_{K2}x_K + v_2 \\ y_3 &= \pi_{13}x_1 + \pi_{23}x_2 + \cdots + \pi_{K3}x_K + v_3 \end{aligned}$$

by least squares and obtain the predicted values

$$\begin{aligned} \hat{y}_2 &= \hat{\pi}_{12}x_1 + \hat{\pi}_{22}x_2 + \cdots + \hat{\pi}_{K2}x_K \\ \hat{y}_3 &= \hat{\pi}_{13}x_1 + \hat{\pi}_{23}x_2 + \cdots + \hat{\pi}_{K3}x_K \end{aligned} \quad (10)$$

- **Step 2:** Replace the endogenous variables, y_2 and y_3 , on the right-hand side of the structural Eqs. 9 by their predicted values from Eqs. 10:

$$\hat{y}_1 = \alpha_2 \hat{y}_2 + \alpha_3 \hat{y}_3 + \beta_1 x_1 + \beta_2 x_2 + e_1^*$$

Estimate the parameters of this equation by least squares

- The properties of the two-stage least squares estimator are:
 - The 2SLS estimator is a biased estimator, but it is consistent
 - In large samples the 2SLS estimator is approximately normally distributed
 - The variances and covariances of the 2SLS estimator are unknown in small samples, but for large samples we have expressions for them that we can use as approximations
 - If you obtain 2SLS estimates by applying two least squares regressions using ordinary least squares regression software, the standard errors and t -values reported in the second regression are not correct for the 2SLS estimator

6 An Example of Two-Stage Least Squares Estimation

Example: Consider a supply and demand model for truffles.

The variables are P = price of truffles, Q = quantity of truffles, PS = price of a substitute for truffles, DI = disposable income of local residents, and PF = price of a factor of production.

Demand:

$$Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_i^d$$

Supply:

$$Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_i^s$$

- The rule for identifying an equation is:

In a system of M equations at least $M - 1$ variables must be omitted from each equation in order for it to be identified

- In the demand equation the variable PF is not included; thus the necessary $M - 1 = 1$ variable is omitted
- In the supply equation both PS and DI are absent; more than enough to satisfy the identification condition

- The reduced-form equations are:

$$\begin{aligned} Q_i &= \pi_{11} + \pi_{21}PS_i + \pi_{31}DI_i + \pi_{41}PF_i + v_{i1} \\ P_i &= \pi_{12} + \pi_{22}PS_i + \pi_{32}DI_i + \pi_{42}PF_i + v_{i1} \end{aligned}$$

- Our estimate of the reduced form for Price is given by:

$$\begin{aligned} \hat{P} &= \hat{\pi}_{12} + \hat{\pi}_{22}PS + \hat{\pi}_{32}DI + \hat{\pi}_{42}PF \\ &= -32.512 + 1.708PS + 7.603DI + 1.354PF \end{aligned}$$

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
<i>C</i>	−4.2795	5.5439	−0.7719	0.4471
<i>P</i>	−0.3745	0.1648	−2.2729	0.0315
<i>PS</i>	1.2960	0.3552	3.6488	0.0012
<i>DI</i>	5.0140	2.2836	2.1957	0.0372

Figure 5: 2SLS Estimates for Truffle Demand

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
<i>C</i>	20.0328	1.2231	16.3785	0.0000
<i>P</i>	0.3380	0.0249	13.5629	0.0000
<i>PF</i>	−1.0009	0.0825	−12.1281	0.0000

Figure 6: 2SLS Estimates for Truffle Supply

R Example

```
library(POE5Rdata)
library(sem) # Structural Equations Models (see function tsls)
data("truffles", package="POE5Rdata")
demand <- tsls(q ~ p + ps + di, ~ ps + di + pf, data = truffles)
supply <- tsls(q ~ p + pf, ~ ps + di + pf, data = truffles)
summary(demand)

##
## 2SLS Estimates
##
## Model Formula: q ~ p + ps + di
##
## Instruments: ~ps + di + pf
##
## Residuals:
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -14.8513  -2.5393   0.9025   0.0000   3.1155   7.5830
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.2794706  5.5438844 -0.77193 0.4471180
## p           -0.3744591  0.1647517 -2.27287 0.0315350 *
## ps           1.2960332  0.3551932  3.64881 0.0011601 **
## di           5.0139771  2.2835559  2.19569 0.0372352 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.92996 on 26 degrees of freedom

summary(supply)

##
## 2SLS Estimates
##
## Model Formula: q ~ p + pf
##
## Instruments: ~ps + di + pf
##
## Residuals:
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -3.7829 -0.8530   0.2270   0.0000   0.7578   3.3475
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 20.0328015  1.22311480 16.37851 1.5543e-15 ***
## p           0.33798157  0.02491956 13.56290 1.4344e-13 ***
## pf          -1.00090937  0.08252794 -12.12813 1.9456e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.4975853 on 27 degrees of freedom
```

7 Supply and Demand at the Fulton Fish Market

- If supply is fixed, with a vertical supply curve, then price is demand-determined
- Higher demand leads to higher prices but no increase in the quantity supplied
- If this is true, then the feedback between prices and quantities is eliminated
- Such models are said to be *recursive* and the demand equation can be estimated by ordinary least squares rather than the more complicated two-stage least squares procedure
- A key point is that “simultaneity” does not require that events occur at a simultaneous moment in time
- Specify the demand equation for this market as:

$$\ln(Q_t) = \alpha_1 + \alpha_2 \ln(PRICE) + \alpha_3 MON_t + \alpha_4 TUE_t + \alpha_5 WED_t + \alpha_6 THU_t + e_i^d \quad (11)$$

where α_2 is the price elasticity of demand

- The supply equation is:

$$\ln(Q_t) = \beta_1 + \beta_2 \ln(PRICE) + \beta_3 STORMY_t + e_i^s \quad (12)$$

where β_2 is the price elasticity of supply

- The necessary condition for an equation to be identified is that in this system of $M = 2$ equations, it must be true that at least $M - 1 = 1$ variable must be omitted from each equation
- In the demand equation the weather variable *STORMY* is omitted, and it does appear in the supply equation
- In the supply equation, the four daily indicator variables that are included in the demand equation are omitted
- The reduced-form equations specify each endogenous variable as a function of all exogenous variables:

$$\ln(Q_t) = \pi_{11} + \pi_{21} MON_t + \pi_{31} TUE_t + \pi_{41} WED_t + \pi_{51} THU_t + \pi_{61} STORMY_t + v_{t1} \quad (13)$$

$$\ln(PRICE_t) = \pi_{12} + \pi_{22} MON_t + \pi_{32} TUE_t + \pi_{42} WED_t + \pi_{52} THU_t + \pi_{62} STORMY_t + v_{t2} \quad (14)$$

- The key reduced-form equation is Eq. 14 for $\ln(PRICE)$:
 - To identify the supply curve, the daily indicator variables must be jointly significant
 - To identify the demand curve, the variable *STORMY* must be statistically significant

The supply has a significant shift variable, so that we can reliably estimate the demand equation

The two-stage least squares estimator performs very poorly if the shift variables are not strongly significant

- To implement two-stage least squares, we take the predicted value from the reduced-form regression and include it in the structural equations in place of the right-hand-side endogenous variable:

$$\hat{\ln}(PRICE) = \hat{\pi}_{12} + \hat{\pi}_{22} MON_t + \hat{\pi}_{32} TUE_t + \hat{\pi}_{42} WED_t + \hat{\pi}_{52} THU_t + \hat{\pi}_{62} STORMY_t$$

- If the coefficients on the daily indicator variables are all identically zero, then:

$$\hat{\ln}(PRICE) = \hat{\pi}_{12} + \hat{\pi}_{62} STORMY_t$$

- If we replace $\ln(PRICE)$ in the supply equation (Eq. 14) with this predicted value, there will be exact collinearity between $\hat{\ln}(PRICE)$ and the variable *STORMY*, which is already in the supply equation
- Two-stage least squares will fail

R Example

```
library(POE5Rdata)
library(systemfit)
data("fultonfish", package="POE5Rdata")
# Demand Equation
D <- lquan~lprice+mon+tue+wed+thu
# Supply Equation
S <- lquan~lprice+stormy
fish.eqs <- list(D, S)
fish.ivs <- ~mon+tue+wed+thu+stormy
fish.sys <- systemfit(fish.eqs, method="2SLS",
                      inst=fish.ivs, data=fultonfish)
summary(fish.sys)
```

```
##
## systemfit results
## method: 2SLS
##
##           N DF      SSR detRCov   OLS-R2 McElroy-R2
## system 222 213 109.612 0.107301 0.094242 -0.597812
##
##           N DF      SSR      MSE      RMSE      R2   Adj R2
## eq1 111 105 52.0903 0.496098 0.704342 0.139124 0.098130
## eq2 111 108 57.5218 0.532610 0.729801 0.049360 0.031755
##
## The covariance matrix of the residuals
##           eq1      eq2
## eq1 0.496098 0.396138
## eq2 0.396138 0.532610
##
## The correlations of the residuals
##           eq1      eq2
## eq1 1.000000 0.770653
## eq2 0.770653 1.000000
##
## 2SLS estimates for 'eq1' (equation 1)
## Model Formula: lquan ~ lprice + mon + tue + wed + thu
## Instruments: ~mon + tue + wed + thu + stormy
##
##           Estimate Std. Error  t value  Pr(>|t|)
```

```

## (Intercept)  8.5059113  0.1661669 51.18896 < 2.22e-16 ***
## lprice      -1.1194169  0.4286450 -2.61152  0.0103334 *
## mon         -0.0254022  0.2147742 -0.11827  0.9060766
## tue         -0.5307694  0.2080001 -2.55177  0.0121574 *
## wed         -0.5663511  0.2127549 -2.66199  0.0089895 **
## thu          0.1092673  0.2087866  0.52334  0.6018373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.704342 on 105 degrees of freedom
## Number of observations: 111 Degrees of Freedom: 105
## SSR: 52.090321 MSE: 0.496098 Root MSE: 0.704342
## Multiple R-Squared: 0.139124 Adjusted R-Squared: 0.09813
##
##
## 2SLS estimates for 'eq2' (equation 2)
## Model Formula: lquan ~ lprice + stormy
## Instruments: ~mon + tue + wed + thu + stormy
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.62835440  0.38897023 22.18256 < 2e-16 ***
## lprice       0.00105931  1.30954697  0.00081  0.99936
## stormy      -0.36324606  0.46491248 -0.78132  0.43632
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.729801 on 108 degrees of freedom
## Number of observations: 111 Degrees of Freedom: 108
## SSR: 57.521843 MSE: 0.53261 Root MSE: 0.729801
## Multiple R-Squared: 0.04936 Adjusted R-Squared: 0.031755

```